CP620 Data Mining Programming — Project Proposal

A Comparative Study of Generative, Discriminative, and Tree-Based Classifiers on the BRFSS Heart Disease Indicators Dataset

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1. Abstract

This project applies the supervised and unsupervised methods covered in CP620 to a binary-classification task on self-reported survey data: predicting whether a respondent has ever reported a history of coronary heart disease or a myocardial infarction, from 21 self-reported health and behavioural indicators. The data come from the U.S. CDC's Behavioral Risk Factor Surveillance System 2015 (BRFSS 2015), accessed via the Kaggle release "Heart Disease Health Indicators Dataset" (253,680 responses, 22 columns). From this source we derive two disjoint (zero-overlap) subsamples - a balanced 10,000-row primary sample (5,000 positive / 5,000 negative) for controlled model comparison, and a stratified 15,000-row secondary imbalanced sample preserving the population's 9.4% positive rate. Three classifiers from distinct method families are implemented from scratch and compared under identical stratified 5-fold cross-validation protocols: a generative Naive Bayes (NB), a discriminative Multilayer Neural Network (MNN) trained by backpropagation, and a Decision Tree (DT) trained by greedy information-gain splitting. Performance is reported with the lecture's metric battery - 2x2 confusion matrix, precision, recall, F1, accuracy, FPR - extended with ROC / AUC and, on the imbalanced sample, Precision-Recall curves and Average Precision (AP). A permutation-importance analysis identifies which health indicators each classifier actually relies on. K-means and hierarchical clustering are run on the unlabelled features as an exploratory step. The deliverable is a focused, fully reproducible comparison study sized appropriately for a class project.

Github: https://github.com/LinklyLuck/CP620-Data-Mining-project

2. Introduction

Cardiovascular disease is a major public-health concern, and this labelled dataset of self-reported heart-disease history paired with health indicators provides a clear binary-classification benchmark. The CDC BRFSS 2015 "Heart Disease Health Indicators" dataset frames this as a 21-feature problem covering chronic-disease status (high blood pressure, high cholesterol, diabetes, stroke), health behaviours (smoking, exercise, diet, alcohol), healthcare access, self-rated general / mental / physical health, and demographics (age, sex, education, income). The setting maps cleanly onto the lecture's running medical-classification examples (the 90%-accurate COVID-19 test, the spam-FREE Naive Bayes worked problem, the cold/allergy/strep confusion matrix). The project answers three questions: (i) which of three classifier families (generative, discriminative, tree-based) gives the best F1 on this dataset; (ii) does the answer change under threshold-independent evaluation (AUC, and on the imbalanced subsample, Average Precision); and (iii) which health indicators each classifier actually relies on. The first question maps to standard course content (NB + MNN); the third extends it with permutation-importance analysis, giving the report a clearly interpretable empirical finding rather than just a metric table.

3. Dataset

CDC BRFSS 2015 Heart Disease Health Indicators, released via Kaggle (Alex Teboul, 2022; originally 253,680 survey responses with 22 columns). Target variable: HeartDiseaseorAttack - whether the respondent has ever been told by a doctor or health professional that they had coronary heart disease or a myocardial infarction. Population class distribution is severely imbalanced (9.4% positive). From this source, two disjoint subsamples are generated by a reproducible sampling procedure (zero source-row overlap; random seed 42; verified by an assertion in the preprocessing script):

- **Primary —** heart_disease_balanced.csv (10,000 rows, 5,000 positive / 5,000 negative). Used for all main model-comparison experiments. Balance lets accuracy remain informative and gives every classifier a fair signal on both classes.
- **Secondary imbalanced sample -** heart_disease_clean.csv (15,000 rows, stratified, preserving the 9.4% positive rate). Used to repeat the evaluation protocol under natural class prevalence. It is drawn first from the raw pool;

the balanced sample is then drawn from remaining source rows. The notebook retains source-row indices only for an explicit zero-overlap check and removes them before modelling.

All 21 features below come from the original BRFSS survey questions; preprocessing on the Kaggle release has already converted them to numeric (no missing values, no string fields). BRFSS rows have no respondent identifier, so any identical feature vectors are retained — they may represent different individuals with the same survey answers.

Feature	Type	Description (BRFSS survey question)
HighBP	binary	Has the respondent been told they have high blood pressure.
HighChol, CholCheck	binary	Has high cholesterol; has had a cholesterol check in the past 5 years.
BMI	continuous	Body Mass Index (12–98 in this dataset).
Smoker, Stroke	binary	Has smoked ≥ 100 cigarettes in lifetime; has ever been told they had a stroke.
Diabetes	ordinal (3)	0 = no diabetes, 1 = pre-diabetes / gestational, 2 = diabetes.
PhysActivity, Fruits, Veggies, HvyAlcoholConsump	binary	Lifestyle: physical activity in past 30 days; eats fruit ≥ 1×/day; eats vegetables ≥ 1×/day; heavy drinker (men > 14 drinks/wk; women > 7 drinks/wk).
AnyHealthcare, NoDocbcCost	binary	Has any healthcare coverage; could not see a doctor in the past 12 months because of cost.
GenHlth	ordinal (5)	Self-rated general health: 1 = excellent, 2 = very good, 3 = good, 4 = fair, 5 = poor.
MentHlth, PhysHlth	count (0–30)	Days of poor mental / physical health in the past 30 days.
DiffWalk	binary	Serious difficulty walking or climbing stairs.
Sex, Age	binary; ordinal (13)	Sex (0 = female, 1 = male); Age bucketed into 13 categories (1 = 18–24, …, 13 = 80+).
Education, Income	ordinal (6, 8)	Education level (1–6, less than elementary → college graduate); Income bracket (1–8, < $10k → ≥ $75k).
target	**binary**	**0 = no heart disease or MI history, 1 = positive history. Project target variable.**

Feature handling: for MNN, all non-binary numeric and ordinal features - BMI, MentHlth, PhysHlth, Diabetes, GenHlth, Age, Education, Income - are z-score standardised using statistics computed from the training fold only. For K-means and hierarchical clustering, the same variables are standardised on the exploratory clustering sample because clustering is not evaluated as a supervised predictor. Decision Tree uses the original numeric values because threshold-based splits do not require scaling. Naive Bayes treats binary and ordinal variables categorically (with Laplace smoothing) and models BMI, MentHlth, and PhysHlth with class-conditional Gaussians.

4. Methods

Three supervised classifiers from distinct methodological families are implemented from scratch in NumPy and compared head-to-head. Two unsupervised methods run on the unlabelled features as exploratory analysis.

4.1 Naïve Bayes (generative)

Applies the Maximum-Posterior Criterion under the conditional-independence assumption:

$$\hat{y} = \arg\max_y \; P(Y=y) \cdot \Pi_i \, P(X_i = x_i \mid Y=y)$$

Categorical features use Laplace-smoothed frequency tables (Lecture 7 procedure); continuous features use class-conditional Gaussians. Independence is unlikely to hold strictly here (e.g., HighBP, GenHlth, and DiffWalk are clearly correlated); the report quantifies the violation via per-pair conditional mutual information and discusses its impact on results.

4.2 Multilayer Neural Network (discriminative)

One hidden layer of H = 16 units, sigmoid activation, sigmoid output, binary cross-entropy loss. Trained by mini-batch (size 16) SGD with momentum ($\beta = 0.9$) and learning-rate decay; early stopping on a 10% internal validation slice; weight initialisation uniform in $[-0.5/\sqrt{d}, +0.5/\sqrt{d}]$. The forward pass and backpropagation (sensitivities δ_k and δ_j) are derived in the report appendix and verified numerically (analytical gradients vs finite-difference gradients agree to 1e-6). All practical-aspects-of-improving-BP items from the MNN lecture (input standardisation, learning-rate schedule, momentum, early stopping) are explicitly implemented.

4.3 Decision Tree (tree-based)

Greedy top-down induction by information-gain splitting (entropy criterion). Continuous features are split on the best threshold via a sorted-value sweep; categorical features by multi-way split. The tree is pre-pruned by two stopping criteria: max depth $\in \{3, 5, 7, \infty\}$ and min samples per leaf ≥ 5. The best (depth, min-leaf) configuration is chosen by inner 3-fold CV on the training fold. Adding DT (a third paradigm) lets the project test whether the NB-vs-MNN comparison is a genuine generative-vs-discriminative difference or just a two-model fluke.

4.4 Unsupervised exploration

K-means (K = 2, 3, 4, 5 with K-means++ and 20 restarts; K chosen by minimising J_W / J_B subject to silhouette ≥ 0.2) is run on the full balanced dataset. Hierarchical agglomerative clustering (average linkage on Euclidean distance) and dendrogram visualisation are run on a stratified random subsample of 1,000 respondents — full-dataset hierarchical clustering at n = 10,000 produces a dendrogram that is computationally feasible but visually unreadable, so the subsample is the right tool here. A dendrogram is produced; cluster IDs are cross-tabulated against the disease label as a sanity check on whether the disease signal is recoverable from the features alone.

5. Evaluation

5.1 Standard metrics (course content)

For each classifier: the 2x2 confusion matrix (rows = actual, columns = predicted, COVID-test layout), plus accuracy, precision, recall, F1, and FPR. Following the Lecture-8 fraud-detection cautionary example (two classifiers, identical accuracy, very different P/R), all metrics are reported side by side rather than collapsed to accuracy. F1 on the positive class is the primary single-threshold summary because it balances precision and recall. Recall is also reported explicitly because failure to identify positive-history respondents is undesirable in this classification setting.

5.2 ROC and AUC (extension)

Each classifier emits a continuous score (NB: log-posterior ratio; MNN: sigmoid output; DT: class-frequency at the predicted leaf). Sweeping the decision threshold tau in [0,1] traces an ROC curve in (FPR, TPR) space. We report both the curve (one figure overlaying all three classifiers) and the area under it (AUC), a threshold-independent summary of ranking performance. ROC / AUC is reported for the balanced primary sample and complements the single-threshold metrics.

5.3 5-fold cross-validation

On the balanced primary sample, stratified 5-fold CV preserves the class ratio in each fold. For each outer fold, all three classifiers are trained on the remaining four folds and predict on the held-out fold. All preprocessing statistics and the Decision Tree's pruning hyperparameters (max-depth, min-samples-per-leaf, selected by inner 3-fold CV) are computed using training-fold data only; the held-out fold is never touched until prediction. The MNN architecture and training hyperparameters are fixed before the final outer-fold run rather than tuned per fold. Confusion-matrix metrics and AUC are reported as mean +/- standard deviation across the five folds.

5.4 Precision-Recall Curve and Average Precision (for the imbalanced subsample)

The same stratified 5-fold CV protocol is repeated independently on the disjoint 15,000-row secondary sample (9.4% positive). Precision-Recall (PR) curves and Average Precision (AP) are computed from pooled out-of-fold prediction scores. Because precision is strongly affected by false positives when the positive class is rare, PR curves and AP are reported alongside ROC-AUC to give a clearer view of positive-class retrieval under imbalance. A no-skill baseline is included to demonstrate that high accuracy can coexist with zero positive-class F1.

5.5 Permutation feature importance (interpretability)

To answer "which features actually drive predictions?", we run permutation-importance analysis on each trained classifier: for each feature x_i, randomly shuffle its values across the test fold (breaking any signal in x_i while preserving its marginal distribution) and measure the resulting F1 drop. Features whose shuffling collapses F1 are important; features whose shuffling does nothing are not. We report the importance ranking for each of the three classifiers and check for agreement — high agreement across three independent classifier families gives high confidence in the ranking; disagreement is itself an interesting finding worth discussing in the report.

6. Pre-stated Hypotheses

- **H1 -** On the balanced 10k primary sample, all three classifiers are expected to achieve useful but imperfect performance; the coarse self-reported indicators should make near-perfect separation unlikely.
- **H2 -** MNN and DT are expected to perform similarly and may outperform NB because correlations among health indicators weaken the conditional-independence assumption used by NB.
- **H3 -** Permutation importance is expected to rank indicators such as GenHlth, Age, HighBP, Diabetes, or BMI highly; any disagreement across classifier families will be reported as a finding rather than forced into agreement.
- **H4 -** On the secondary imbalanced sample (9.4% positive), an all-negative baseline achieves about 0.906 accuracy but positive-class F1 = 0; AP for a no-skill ranking baseline is approximately the positive prevalence (0.094).
- **H5 -** K-means is expected to show moderate rather than near-perfect agreement with the target labels because heart-disease history is unlikely to be the dominant structure in the survey-feature space.

If a hypothesis is contradicted by the data, the report will state so directly and analyse the cause; pre-stating hypotheses prevents post-hoc storytelling.

7. Timeline and Deliverables

Week	Phase	Deliverable
1	Data + EDA	Download dataset, preprocess, freeze CSV; per-feature histograms; class distribution; unsupervised K-means + hierarchical clustering with dendrogram.
2	NB + DT	Implement NB and DT from scratch in NumPy; unit-test NB against the buys_computer worked example; unit-test DT against a hand-computed information-gain split.
3	MNN	Implement MNN forward + backpropagation; numerical gradient check; train a baseline; freeze H and learning-rate settings before the final outer-fold evaluation.
4	Evaluation	Run stratified 5-fold CV on both samples; confusion matrices; P/R/F1; ROC + AUC; PR curves + AP on the imbalanced sample; permutation feature importance.
5	Analysis + writing	Compare results against pre-stated hypotheses; analyse imbalanced-sample errors; interpret permutation importance; draft report.
6	Polish + presentation	Final 12–15-page report; 10-slide presentation; single-command-reproducible Jupyter notebook regenerating every figure and table.

8. Risks and Mitigations

Risk	Mitigation
Severe class imbalance on the stratified 15k subsample (9.4% positive) collapses naïve classifiers to the majority class.	Repeat the same stratified 5-fold evaluation on the imbalanced sample; report positive-class P/R/F1 and AP; retain empirical class priors for NB; examine class-weighted MNN/DT only as optional mitigation analysis.
MNN training across 10k × 22-feature × 5-fold CV could be slow.	Mini-batch training with early stopping; small hidden layer (H = 16); each fold trains in under a minute on a laptop CPU. Total compute budget for all classifiers and folds is < 30 minutes.
Naïve-independence assumption violated by correlated features (e.g., HighBP / GenHlth / DiffWalk).	Quantify the violation empirically via conditional mutual information; report and discuss; DT and MNN serve as independence-free comparison points.
Decision Tree overfits without pruning; cross-classifier comparison fairness.	Inner 3-fold CV on the training fold selects DT max-depth and min-samples-per-leaf. MNN hyperparameters fixed before outer-fold evaluation (no per-fold tuning) to keep the CV protocol clean and avoid test-fold leakage; the report acknowledges this leaves small headroom on MNN.
Results diverge from pre-stated hypotheses.	Treat divergence as a finding, not a failure: report it honestly and analyse its cause.

9. Summary

This project takes the CDC BRFSS 2015 Heart Disease Health Indicators dataset (253,680 survey responses) and derives two disjoint subsamples: a balanced 10,000-row primary sample for controlled model comparison and a stratified 15,000-row secondary sample preserving the population's 9.4% positive rate. Three classifier families - generative (Naive Bayes), discriminative (Multilayer NN), and tree-based (Decision Tree) - are implemented from scratch in NumPy and evaluated under separate stratified 5-fold protocols. Evaluation uses the course metric battery (confusion matrix, P/R/F1, accuracy, FPR), ROC / AUC, and Precision-Recall curves with Average Precision on the imbalanced sample. Permutation importance identifies which health indicators each classifier relies on, while K-means and hierarchical clustering provide exploratory structure analysis. The deliverable is a focused, reproducible classification study that demonstrates both breadth and depth under a clean two-regime evaluation design.